Fundamental Calculus/Differential and Integral Calculus

Prepared By : Nasrina Parvin

Lecturer, INS, UIU

Reference : Calculus 10-th Edition by Howard Anton, Irl Bivens and Stephen Davis

Function: Domain and Range

- Section: 0.1

Vertical line test(Page -5)

0.1.3 THE VERTICAL LINE TEST *A curve in the xy-plane is the graph of some function f if and only if no vertical line intersects the curve more than once.*

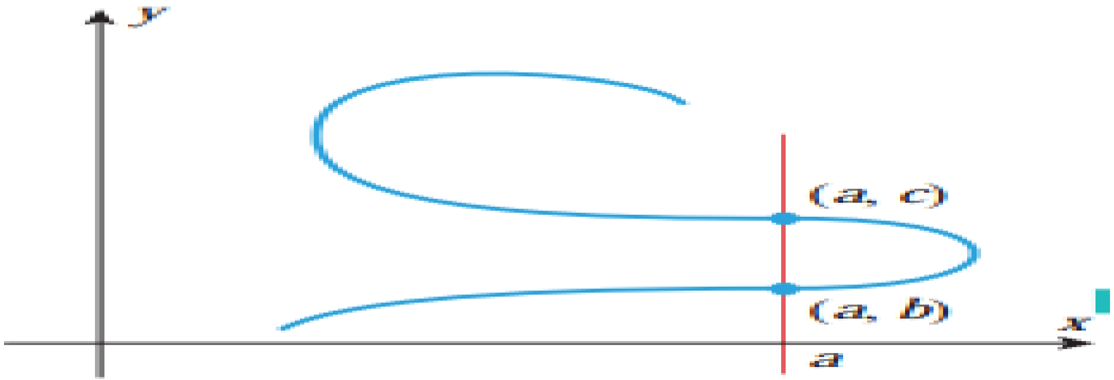
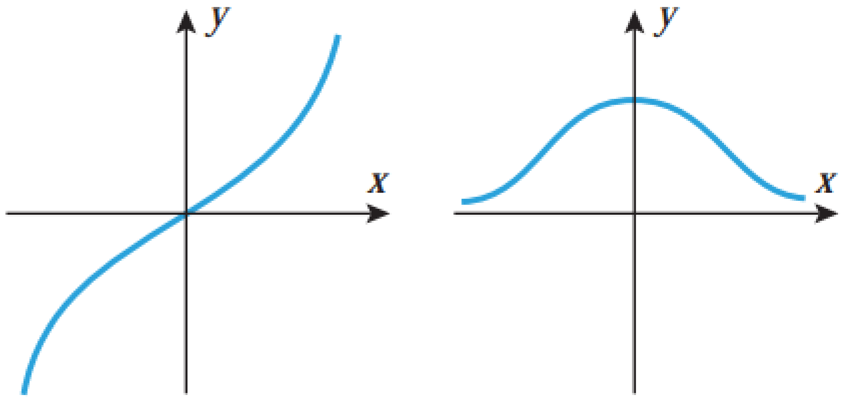


Figure 0.1.7 This curve cannot be the graph of a function.

Vertical line test(Page-12)

3. In each part of the accompanying figure, determine whether the graph defines y as a function of x.

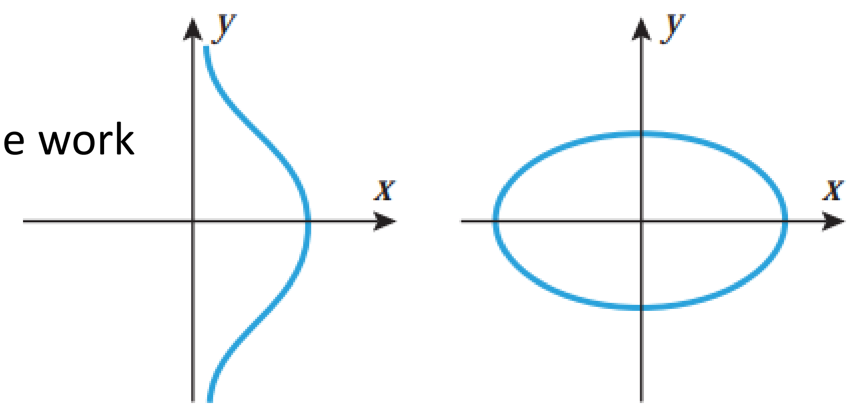


(*a*) (*b*)

Home work

Home work

Determine whether the given curve is function or not.

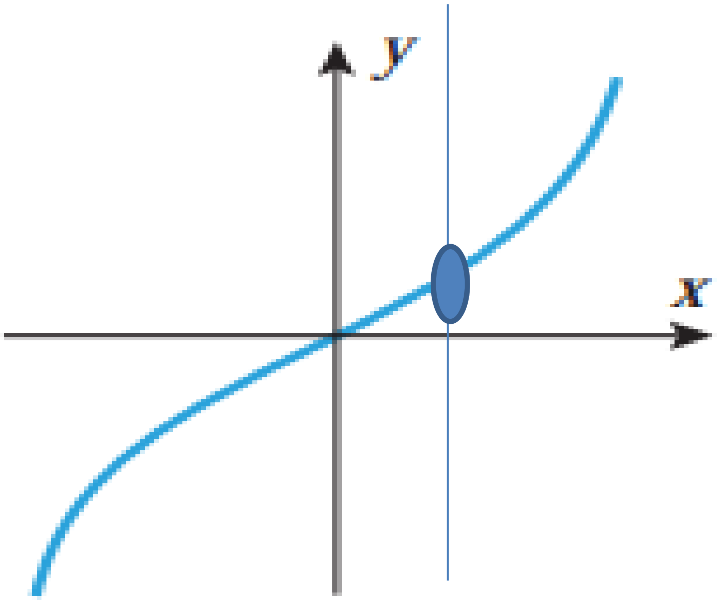


(*c*) (*d*)

Figure Ex-3

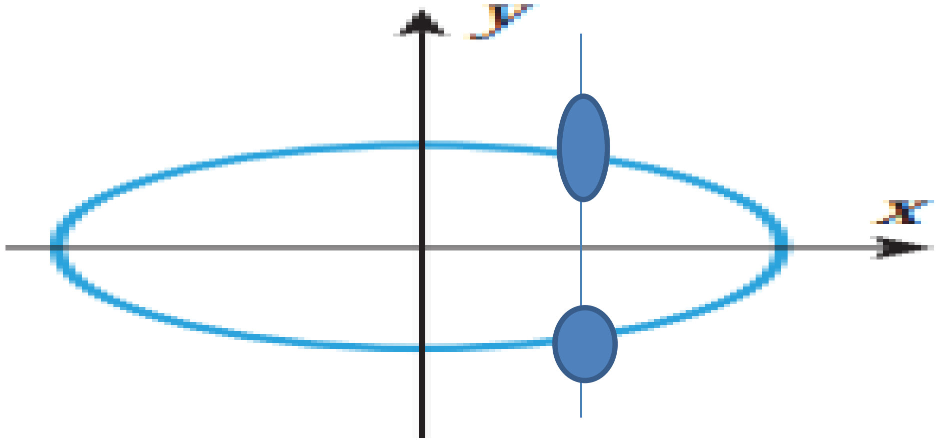
Solution: 3(a):

From Vertical line test we can say that, the vertical line cuts or intersects the graph only one point, so the given curve is a graph of function.



(a)

- <u>Solution: 3(d):</u>
- From Vertical line test we can say that, the vertical line cuts or intersects the graph more than one point, so the given curve is not a graph of function.



Function

7. Find $f(0)$, $f(2)$, $f(-2)$, $f(3)$, $f(\sqrt{2}\,)$, and $f(3t)$.

(a) $f(x) = 3x^2 - 2$ (b) $f(x) = \begin{cases} \dfrac{1}{x}, & x > 3 \\ 2x, & x \le 3 \end{cases}$

8. Find $g(3)$, $g(-1)$, $g(\pi)$, $g(-1.1)$, and $g(t^2 - 1)$.

(a) $g(x) = \dfrac{x+1}{x-1}$ (b) $g(x) = \begin{cases} \sqrt{x+1}, & x \ge 1 \\ 3, & x < 1 \end{cases}$

Function

7. **(a)** $f(0) = 3(0)^2 - 2 = -2$; $f(2) = 3(2)^2 - 2 = 10$; $f(-2) = 3(-2)^2 - 2 = 10$; $f(3) = 3(3)^2 - 2 = 25$; $f(\sqrt{2}) = 3(\sqrt{2})^2 - 2 = 4$; $f(3t) = 3(3t)^2 - 2 = 27t^2 - 2$.

(b) $f(0) = 2(0) = 0$; $f(2) = 2(2) = 4$; $f(-2) = 2(-2) = -4$; $f(3) = 2(3) = 6$; $f(\sqrt{2}) = 2\sqrt{2}$; $f(3t) = 1/(3t)$ for $t > 1$ and $f(3t) = 6t$ for $t \leq 1$.

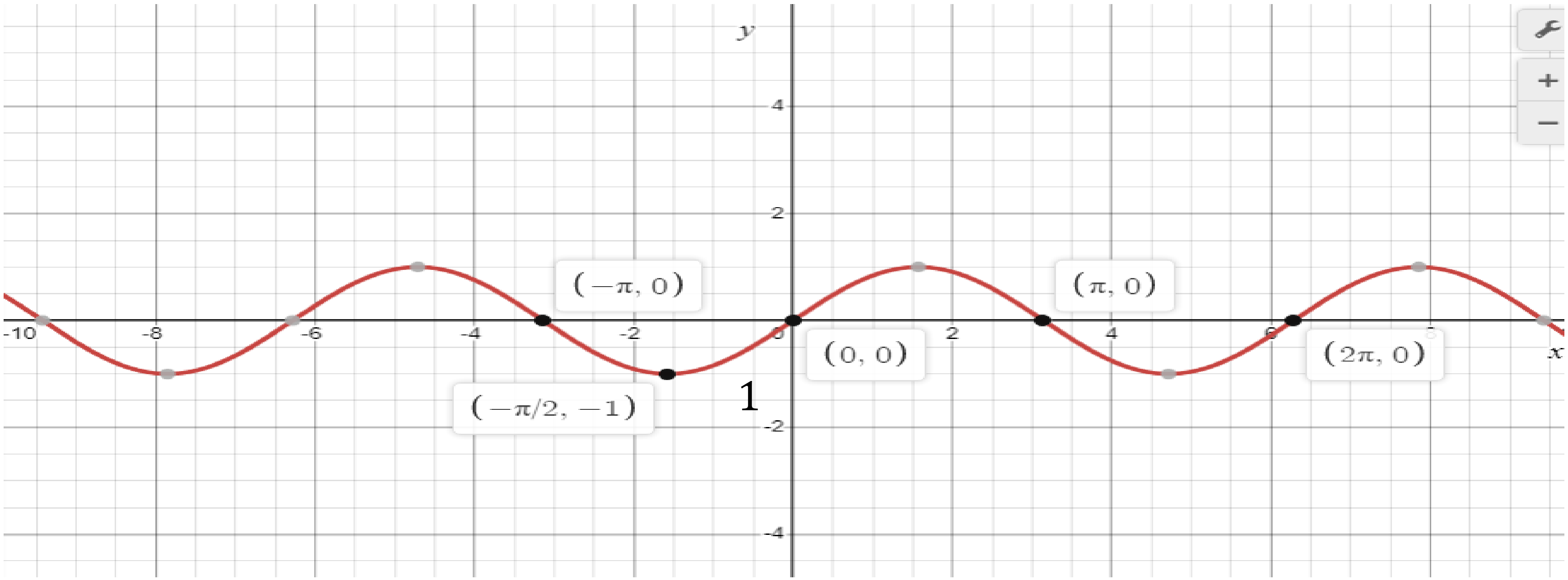
8. **(a)** $g(3) = \dfrac{3+1}{3-1} = 2$; $g(-1) = \dfrac{-1+1}{-1-1} = 0$; $g(\pi) = \dfrac{\pi+1}{\pi-1}$; $g(-1.1) = \dfrac{-1.1+1}{-1.1-1} = \dfrac{-0.1}{-2.1} = \dfrac{1}{21}$; $g(t^2-1) = \dfrac{t^2-1+1}{t^2-1-1} = \dfrac{t^2}{t^2-2}$.

(b) $g(3) = \sqrt{3+1} = 2$; $g(-1) = 3$; $g(\pi) = \sqrt{\pi+1}$; $g(-1.1) = 3$; $g(t^2-1) = 3$ if $t^2 < 2$ and $g(t^2-1) = \sqrt{t^2-1+1} = |t|$ if $t^2 \geq 2$.

Graph of $y = |x|$

Domain is $\mathbb{R}$ $or(-\infty, +\infty)$ and range is $[0\ , +\infty)$

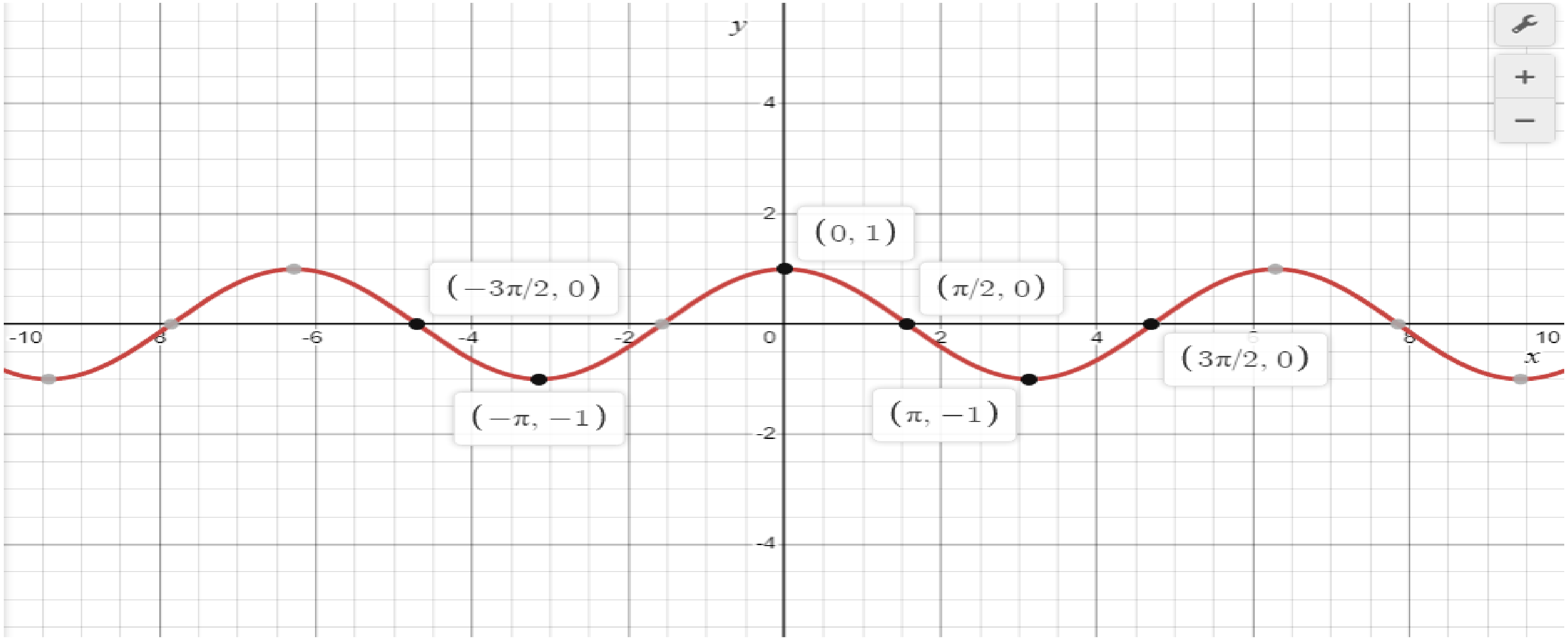
Trigonometric Function



$y = sinx$

Domain: (-∞, +∞) $and\ \ Range$: [-1,1]

Trigonometric Function



$y = Cos x$

Domain: (-∞, +∞) $and \ \ Range$: [-1,1]

Chapter : 0.2(Page-15)

NEW FUNCTIONS FROM OLD

Intercepts

- If we put $x = 0$, we get y intercepts, if we put
- y$= 0$, we get x intercepts.

Find the intercepts of $f(x) = x^2 - 4x$. If we put y$= 0$, y$= f(x) = x^2 - 4x = 0$, if $\mathrm{y} = 0$

$$x(x-4) = 0$$

$\Rightarrow x = 0, x = 4, \text{so x intercepts are } (\mathrm{x}, \mathrm{y}) = (0{,}0) \text{and } (4{,}0)$, If we put x$= 0$, y$= f(0) = 0^2 - 4.0 = 0$, so y intercepts are $(\mathrm{x}, \mathrm{y}) = (0{,}0)$

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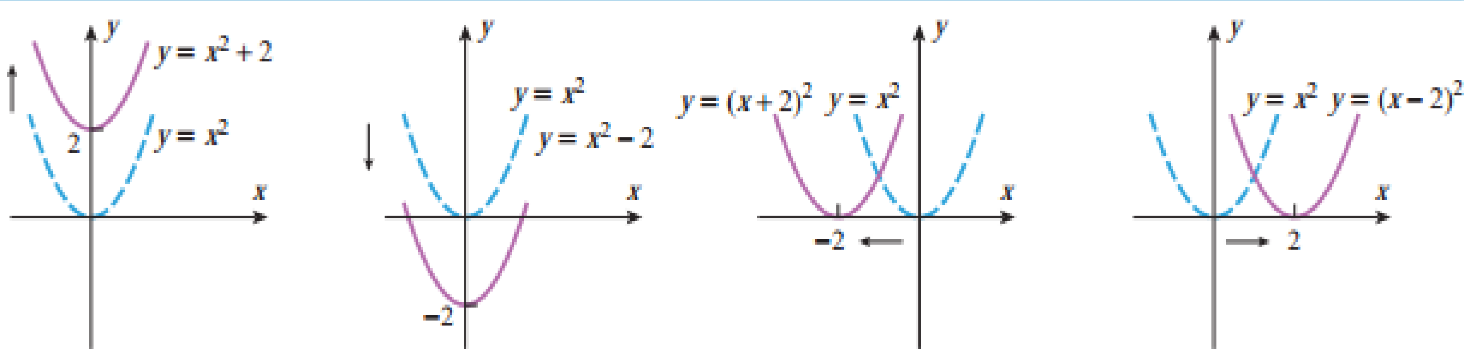
0.2 NEW FUNCTIONS FROM OLD

TRANSLATIONS (Page-20) :

Table 0.2.2
TRANSLATION PRINCIPLES

OPERATION ON $y = f(x)$	Add a positive constant c to $f(x)$	Subtract a positive constant c from $f(x)$	Add a positive constant c to x	Subtract a positive constant c from x
NEW EQUATION	$y = f(x) + c$	$y = f(x) - c$	$y = f(x + c)$	$y = f(x - c)$
GEOMETRIC EFFECT	Translates the graph of $y = f(x)$ up c units	Translates the graph of $y = f(x)$ down c units	Translates the graph of $y = f(x)$ left c units	Translates the graph of $y = f(x)$ right c units

EXAMPLE



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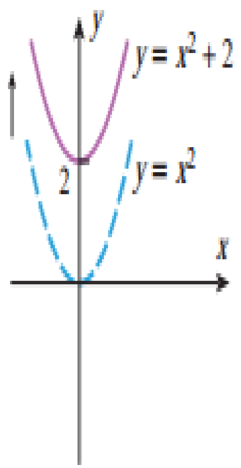
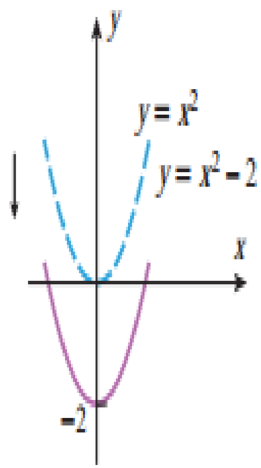
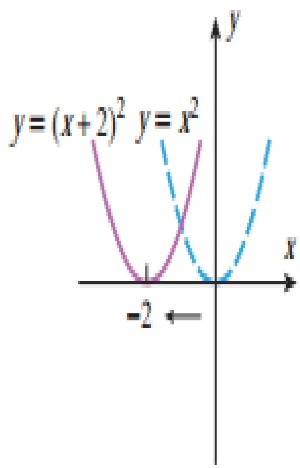
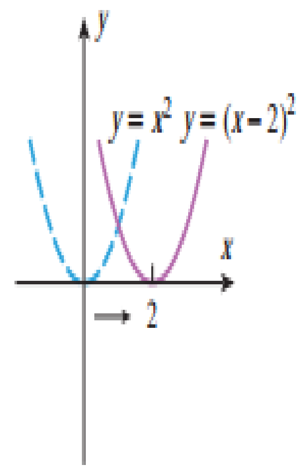
NEW FUNCTIONS FROM OLD

- Table 0.2.2 illustrates the geometric effect on the graph of $y = f(x)$ of adding or subtracting a *positive* constant c to f or to its independent variable x. For example, the first result in the table illustrates that adding a positive constant c to a function f adds c to each y-coordinate of its graph, thereby shifting the graph of f up by c units.
- Similarly, subtracting c from f shifts the graph down by c units. On the other hand, if a positive constant c is added to x, then the value of $y = f(x + c)$ at $x - c$ is $f(x)$; and since the point $x - c$ is c units to the left of x on the x-axis, the graph of $y = f(x + c)$ must be the graph of $y = f(x)$ shifted left by c units.
- Similarly, subtracting c from x shifts the graph of $y = f(x)$ right by c units.

Table 0.2.2

TRANSLATION PRINCIPLES

OPERATION ON $y = f(x)$	Add a positive constant c to $f(x)$	Subtract a positive constant c from $f(x)$	Add a positive constant c to x	Subtract a positive constant c from x
NEW EQUATION	$y = f(x) + c$	$y = f(x) - c$	$y = f(x + c)$	$y = f(x - c)$
GEOMETRIC EFFECT	Translates the graph of $y = f(x)$ up c units	Translates the graph of $y = f(x)$ down c units	Translates the graph of $y = f(x)$ left c units	Translates the graph of $y = f(x)$ right c units
EXAMPLE				

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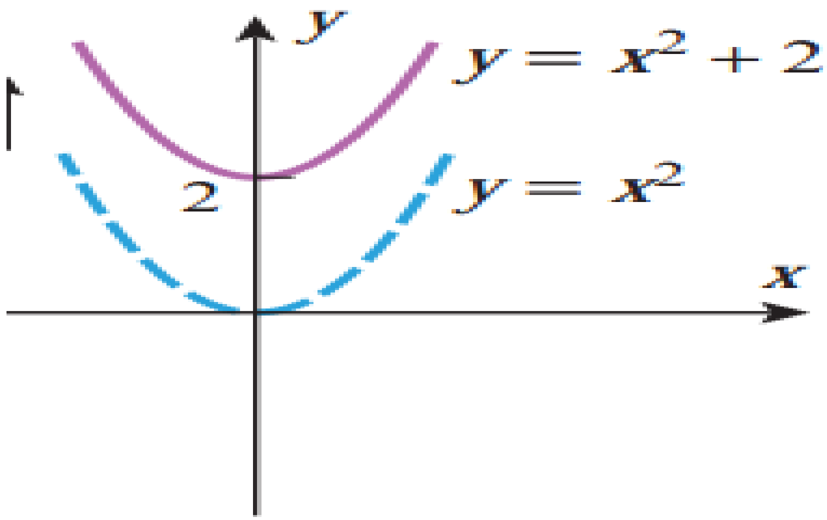
Draw the graphs. Also find the domain and range of the functions: (Page:20-21)

- $f(x) = x^2 + 2$
- $f(x) = x^2 - 2$
- $f(x) = (x+2)^2$
- $f(x) = (x-2)^2$
- $f(x) = (x-2)^2 + 1$
- $f(x) = \sqrt{x-3}$
- $f(x) = \sqrt{x+3}$

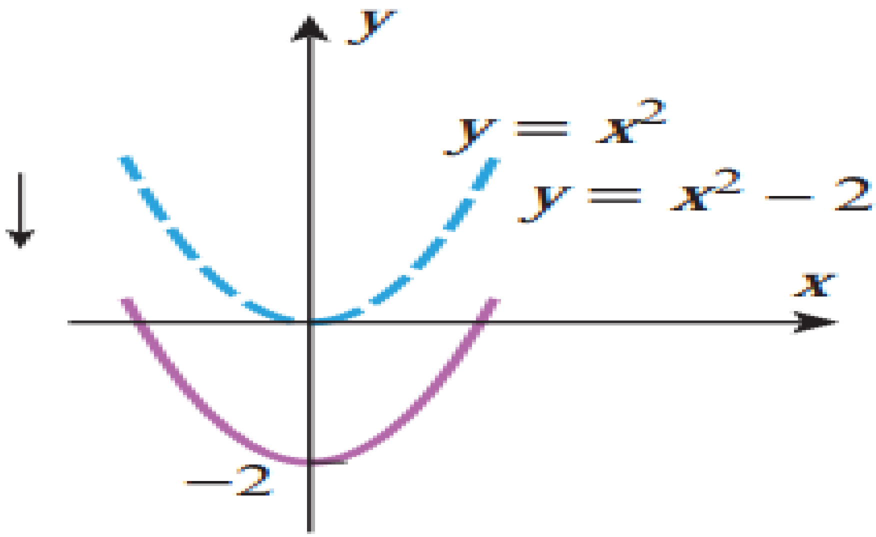
Finding domain and range of $y = f(x) = x^2$+2



Translation of y axis⟶ y intercepts

$Domain\ is$ $(-\infty, +\infty)$ or $\mathbb{R}$, and Range is $[2, +\infty)$

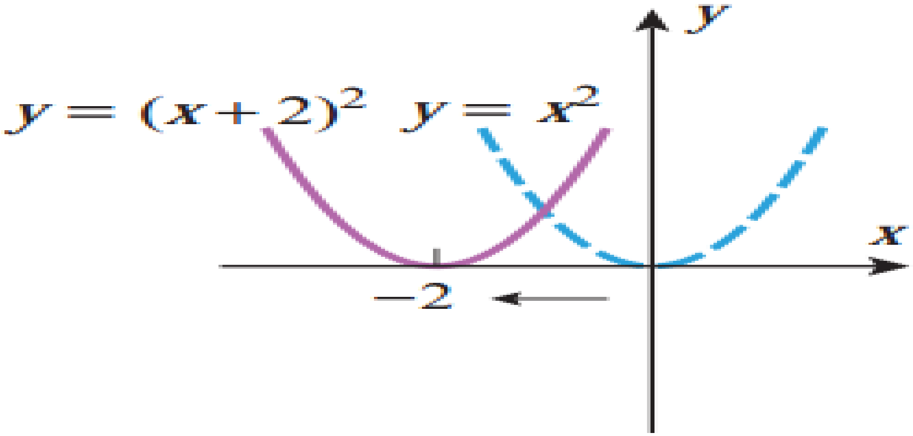
Finding domain and range of $y = f(x) = x^2 - 2$



Translation of y axis$\longrightarrow$ y intercepts

$Domain\ is$ $(-\infty, +\infty)$ or $\mathbb{R}$, and Range is $[-2, +\infty)$

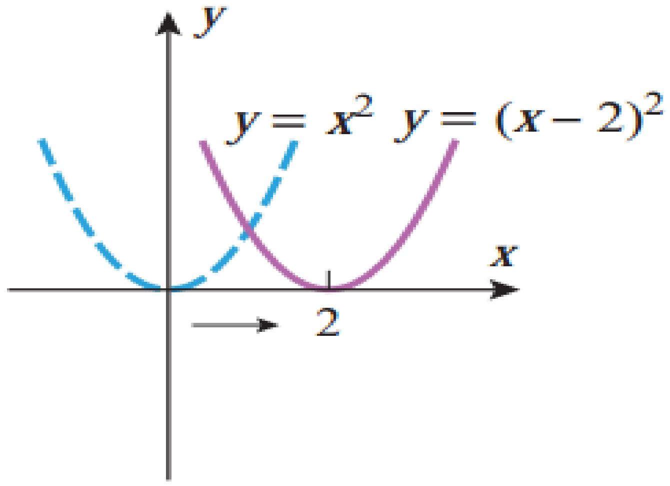
Finding domain and range of $y = f(x) = (x+2)^2$



Translation of x axis$\longrightarrow$ X intercepts

$Domain\ is\ (-\infty, +\infty)$ or $\mathbb{R}$, and Range $is\ [0, +\infty)$

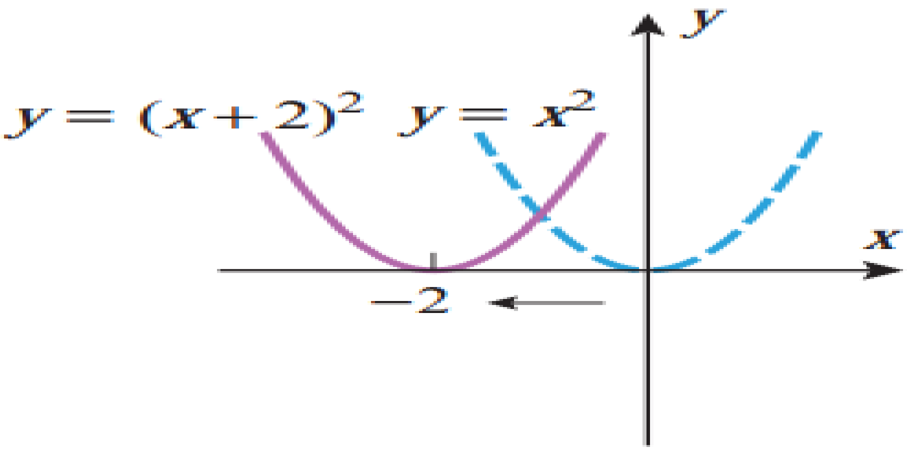
Finding domain and range of $y = f(x) = (x-2)^2$



Translation of x axis⟶ X intercepts

$Domain\ is$ $(-\infty, +\infty)$ or $\mathbb{R}$, and Range is $[0, +\infty)$

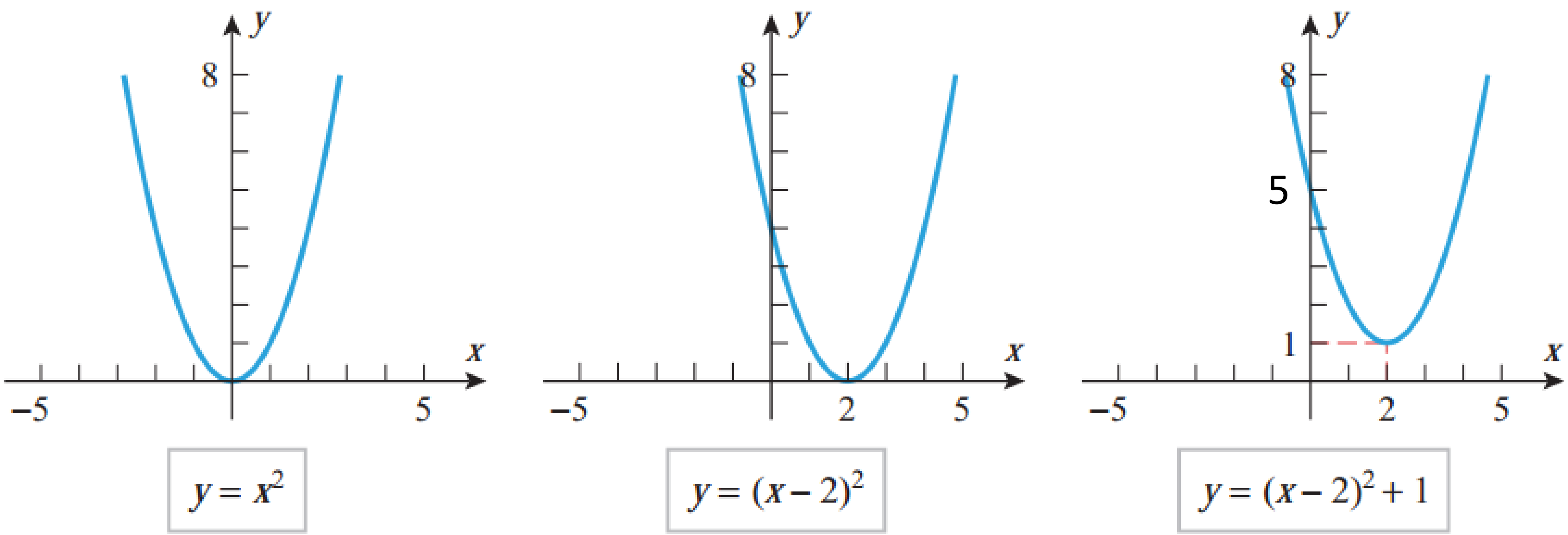
Finding domain and range of $y = f(x) = (x-2)^2$



Translation of x axis$\longrightarrow$ X intercepts

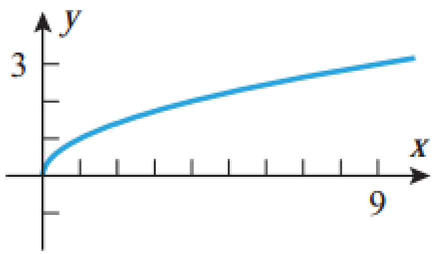
$Domain\ is\ (-\infty, +\infty)$ or $\mathbb{R}$, and Range $is\ [0, +\infty)$

Finding domain and range of $y = f(x) = (x-2)^2 + 1$

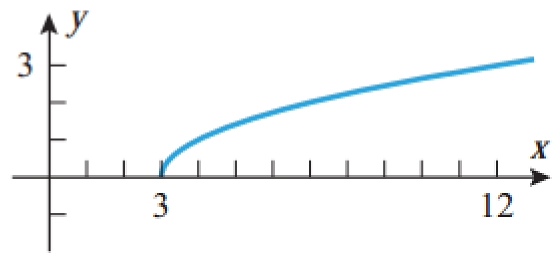


$Domain\ is\ (-\infty, +\infty)$ or $\mathbb{R}$, and Range $is\ [1, +\infty)$

Finding domain and range of $y = f(x) = \sqrt{x-3}$



$y = \sqrt{x}$

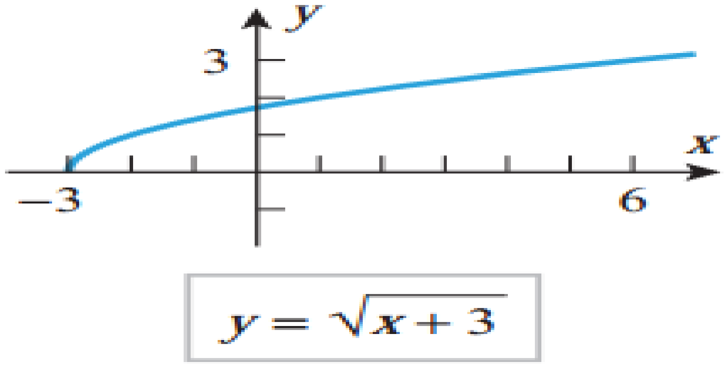


$y = \sqrt{x-3}$

Translation of x axis⟶ X intercepts

$Domain\ is\ [3, +\infty)$ and Range $is\ [0, +\infty)$

Finding domain and range of $y = f(x) = \sqrt{x+3}$



Translation of x axis$\longrightarrow$ X intercepts

$Domain\ is\ [-3, +\infty)$ and Range $is\ [0, +\infty)$

Intercepts

- If we put $x = 0$, we get y intercepts, if we put
- y$= 0$, we get x intercepts.

Find the intercepts of $f(x) = x^2 - 4x$. If we put y$= 0$, y$= f(x) = x^2 - 4x = 0$, if $y = 0$

$$x(x-4) = 0$$

$\Rightarrow x = 0, x = 4, \text{so x intercepts are } (\text{x}, \text{y}) = (0,0) \text{and } (4,0)$, If we put x$= 0$, y$= f(0) = 0^2 - 4.0 = 0, \text{so y intercepts are } (\text{x}, \text{y}) = (0,0)$

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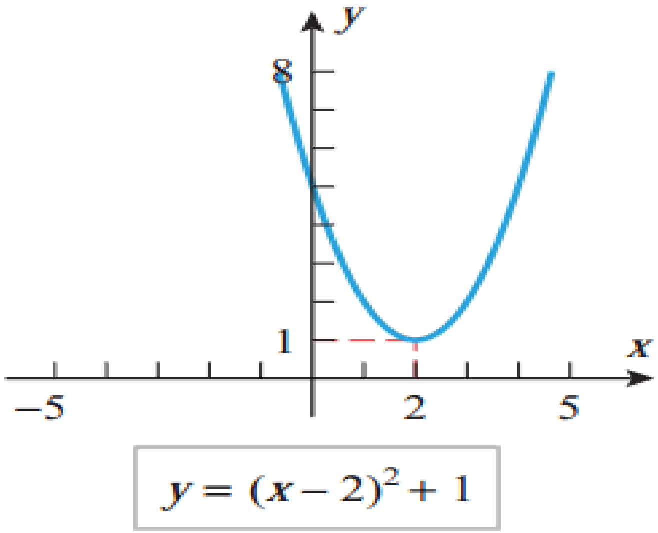
Draw the graph of $f(x) = x^2 - 4x + 5$,

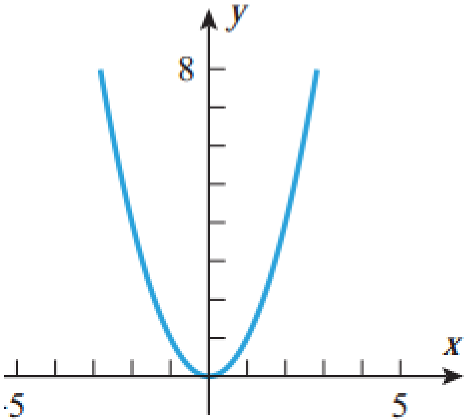
- <u>Example 8:</u>

Now,

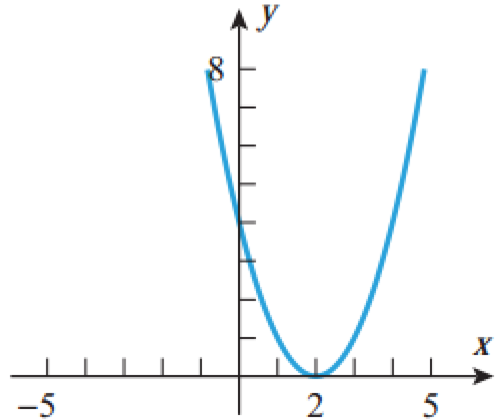
$$f(x) = x^2 - 4x + 5 = x^2 - 2.x.2 + 2^2 + 1$$
$$= (x-2)^2 + 1$$



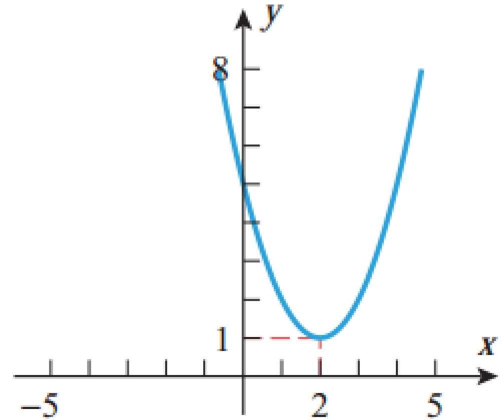
Translations



$y = x^2$



$y = (x-2)^2$



$y = (x-2)^2 + 1$

Sketch the graph of $f(x) = x^2 - 4x$, also find their domain and range.

- Now, $f(x) = x^2 - 4x = x^2 - 2.x.2 + 2^2 - 4 = (x-2)^2 - 4$

If we put y= 0, y= $f(x) = x^2 - 4x = 0$, if $\mathrm{y} = 0$

$$x(x-4) = 0$$

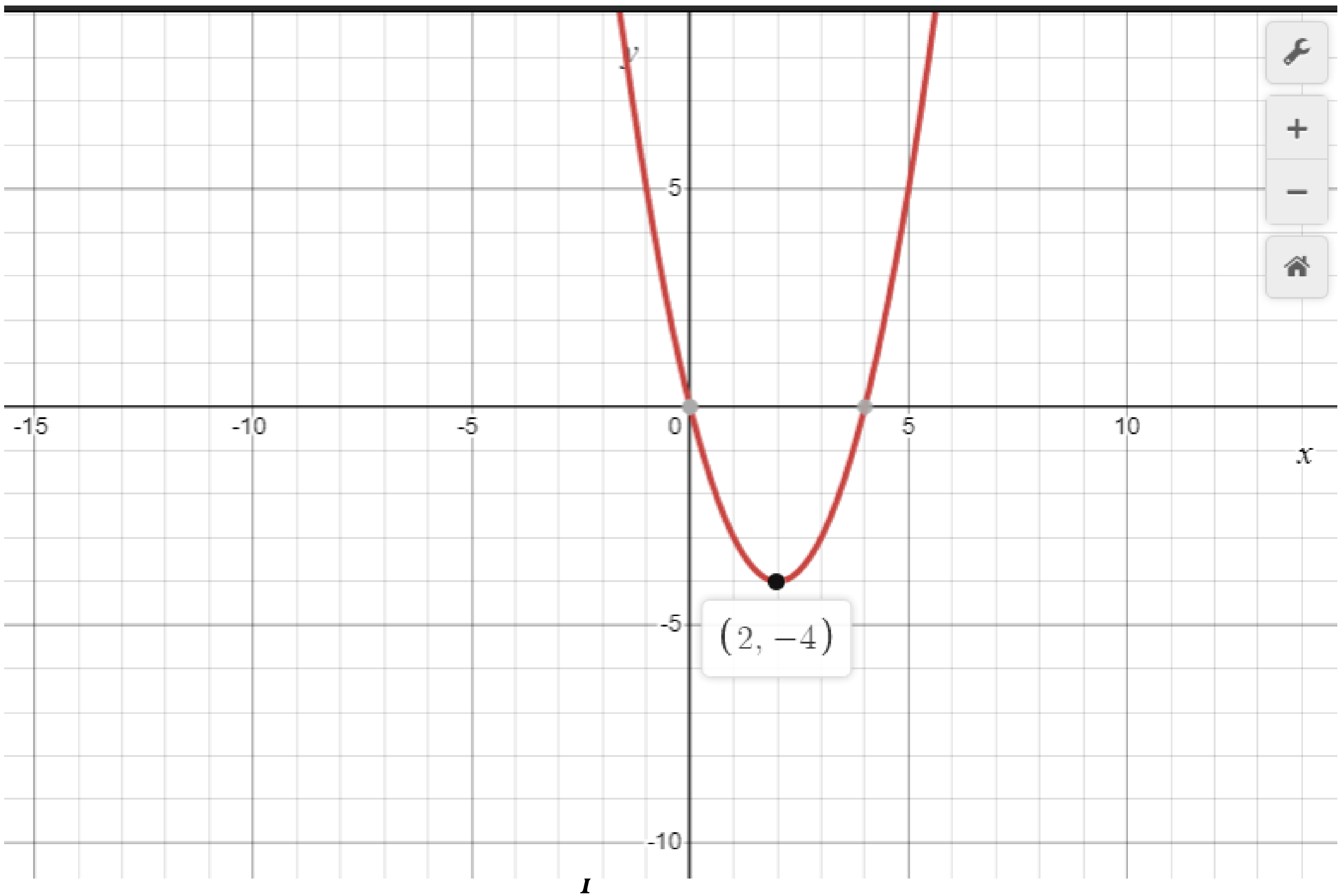
$\Rightarrow x = 0, x = 4$, so x intercepts are $(\mathrm{x}, \mathrm{y}) = (0,0)$and $(4,0)$, If we put x= 0, y= $f(0) = 0^2 - 4.0 = 0$, so x intercepts are $(\mathrm{x}, \mathrm{y}) = (0,0)$

If we put y= 0, y= $f(x) = x^2 - 4x = 0$, if $\mathrm{y} = 0$

$$x(x-4) = 0$$

$\Rightarrow x = 0, x = 4$, so x intercepts are $(\mathrm{x}, \mathrm{y}) = (0,0)$and $(4,0)$, If we put x= 0, y= $f(0) = 0^2 - 4.0 = 0$, so y intercepts are $(\mathrm{x}, \mathrm{y}) = (0,0)$

Graph of $f(x) = x^2 - 4x$



REF LECTIONS

- REF LECTIONS(Page-21):
- The graph of $y = f(-x)$ is the reflection of the graph of $y = f(x)$ about the y-axis because the point (x, y) on the graph of $f(x)$ is replaced by $(-x, y)$. Similarly, the graph of $y = -f(x)$ is the reflection of the graph of $y = f(x)$ about the x-axis because the point (x, y) on the graph of $f(x)$ is replaced by $(x, -y)$ [the equation $y = -f(x)$ is equivalent to $-y = f(x)$]. This is summarized in Table 0.2.3.

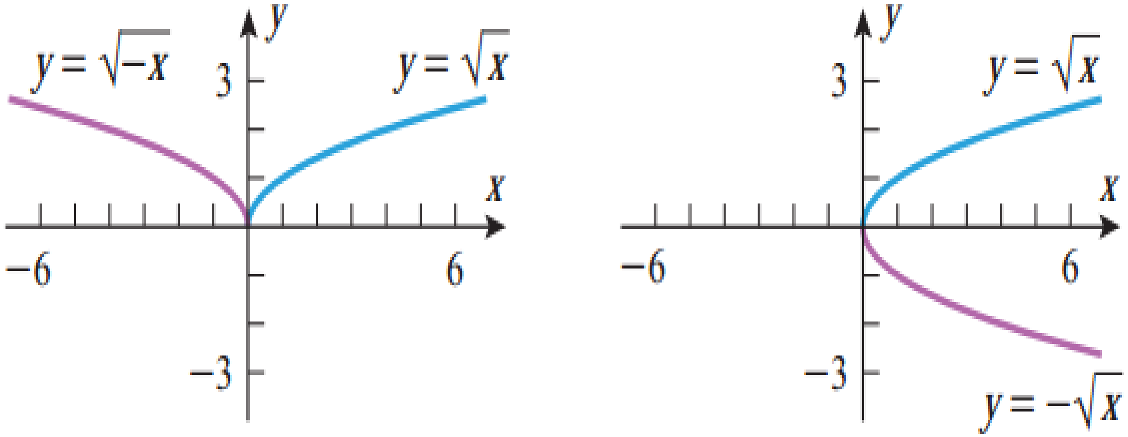
REFLECTIONS

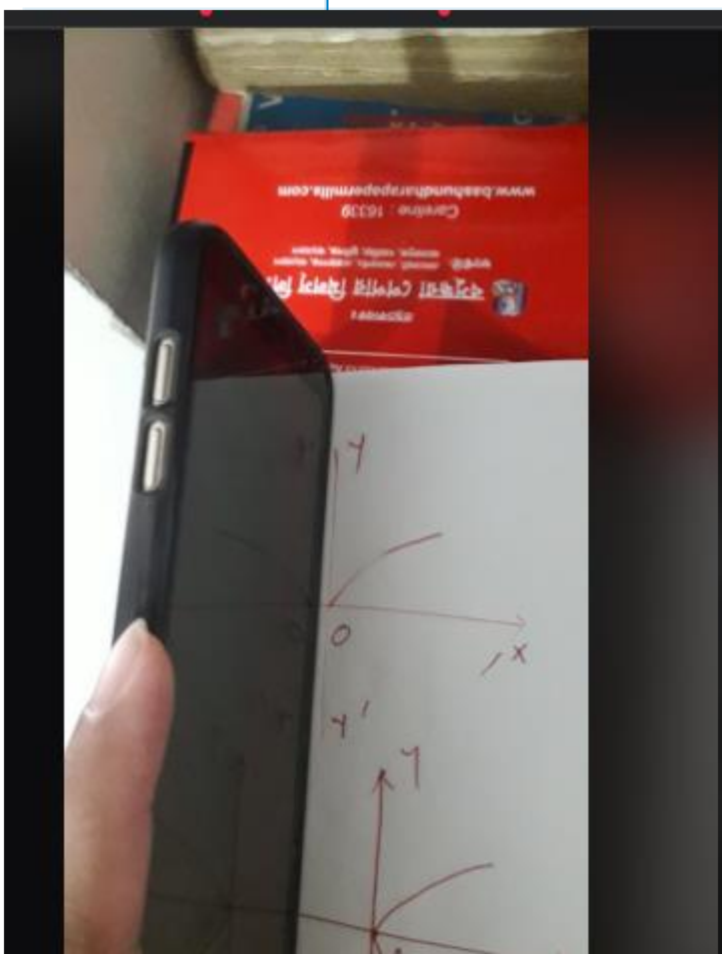
Table 0.2.3

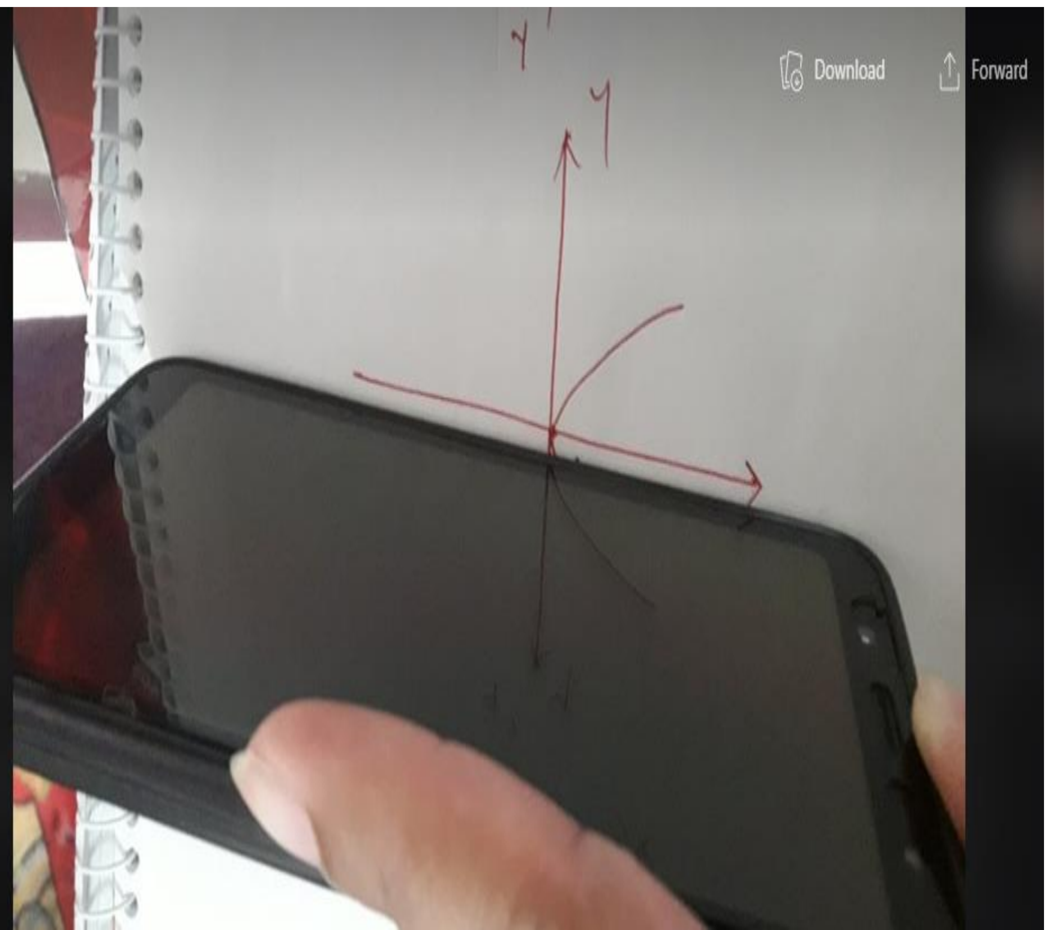
REFLECTION PRINCIPLES

OPERATION ON $y = f(x)$	Replace x by $-x$	Multiply $f(x)$ by -1
NEW EQUATION	$y = f(-x)$	$y = -f(x)$
GEOMETRIC EFFECT	Reflects the graph of $y = f(x)$ about the y-axis	Reflects the graph of $y = f(x)$ about the x-axis

EXAMPLE



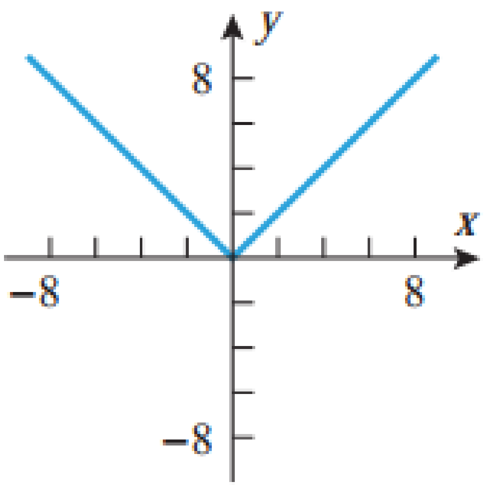




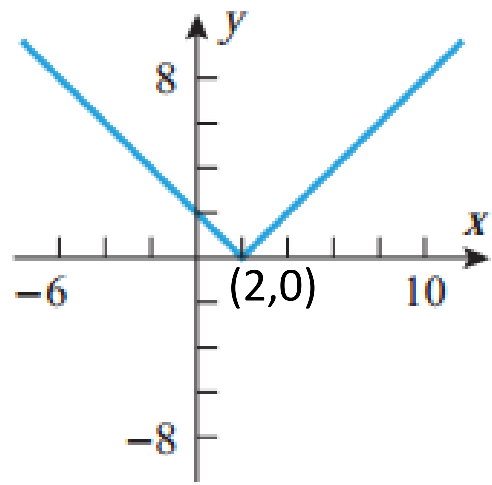
Graph of $y = |x|$

Domain is $\mathbb{R}\ or (-\infty, +\infty)$ and range is $[0\ ,+\infty)$

<u>Example :</u> Sketch the graph of $f(x)$=$y = |x - 2|$



$y = |x|$



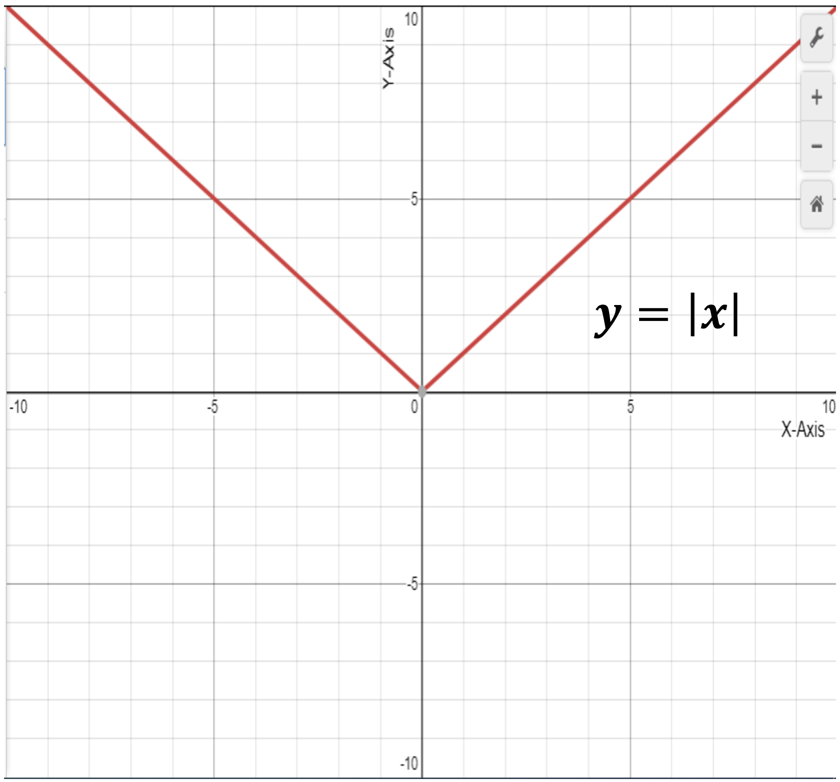
$y = |x - 2|$

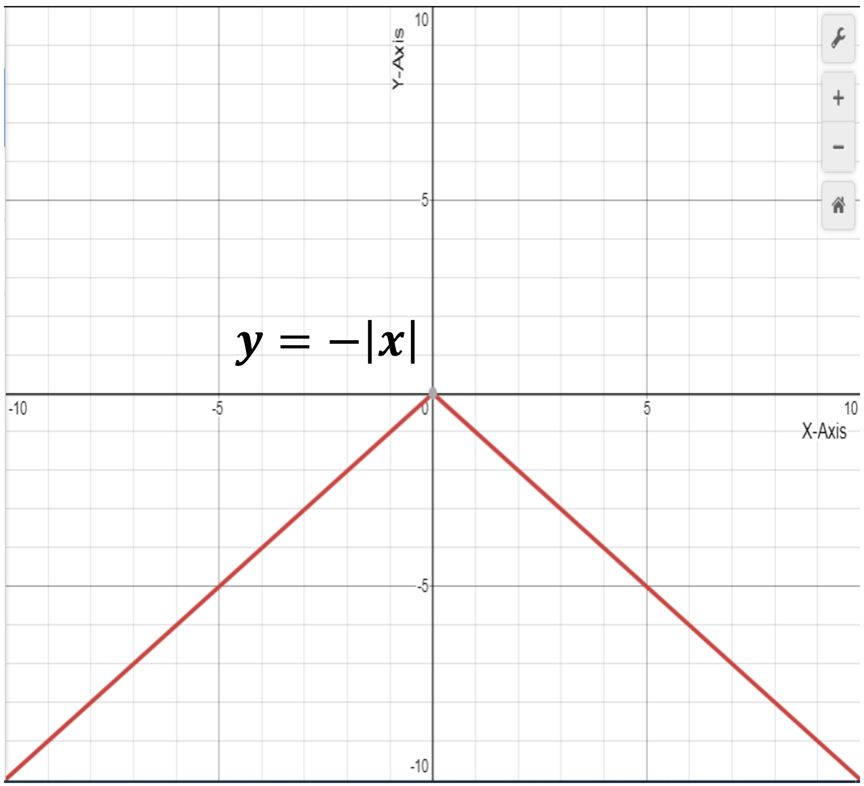
Sketch the graph of $f(x)$=$y = |x - 2|$ +4 [Home work]

Domain is $\mathbb{R}$ $or(-\infty, +\infty)$ and range is $[0, +\infty)$

Example 10(Page-21):

Sketch the graph of $f(x)$=$y = 4 - |x - 2|$

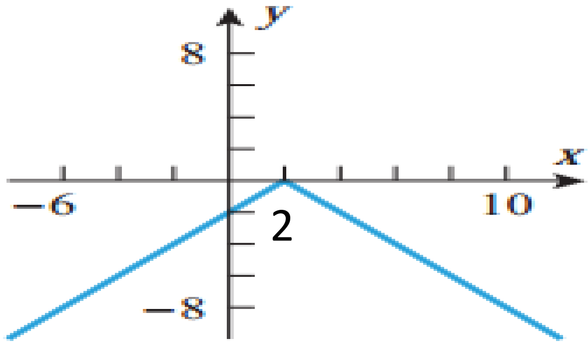




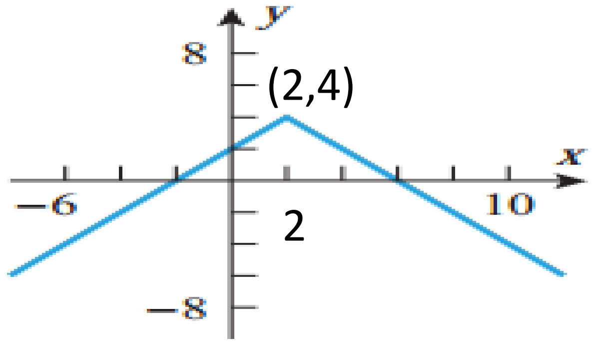
Domain is $\mathbb{R}$ $or(-\infty, +\infty)$ and range is (0,+∞]

Domain is $\mathbb{R}$ $or(-\infty, +\infty)$ and range is (-∞, 0]

Graph of $f(x)$=$y = 4 - |x - 2|$



$y = -|x - 2|$



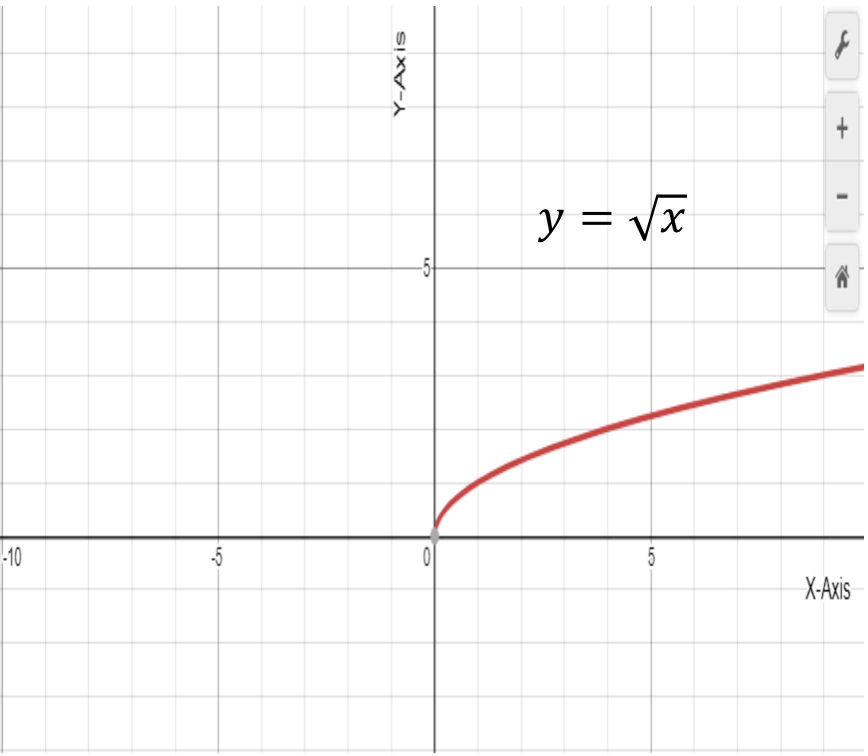
$y = 4 - |x - 2|$

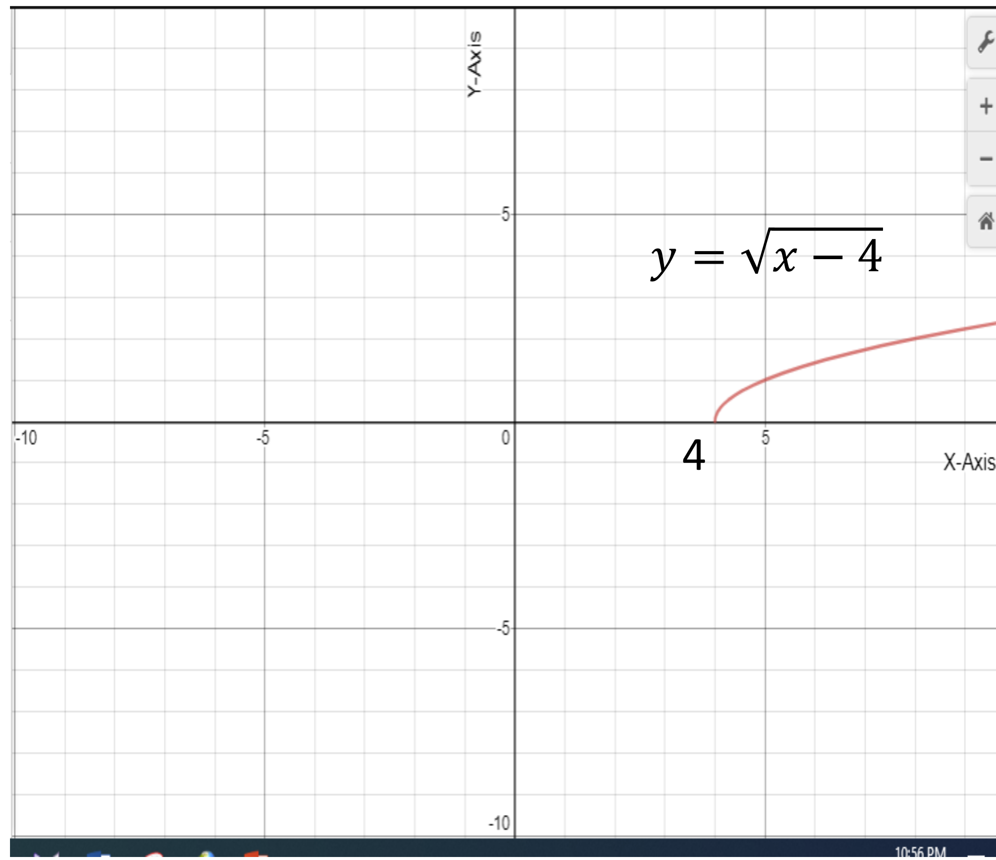
Domain is $\mathbb{R}$ $or(-\infty, +\infty)$ and range is (-∞, $4]$

main is $\mathbb{R}$ $or(-\infty, +\infty)$ and range is (-∞, $0]$

Exercise: 10(Page-21):

Sketch the graph of $f(x)$=$y = 1 + \sqrt{x-4}$

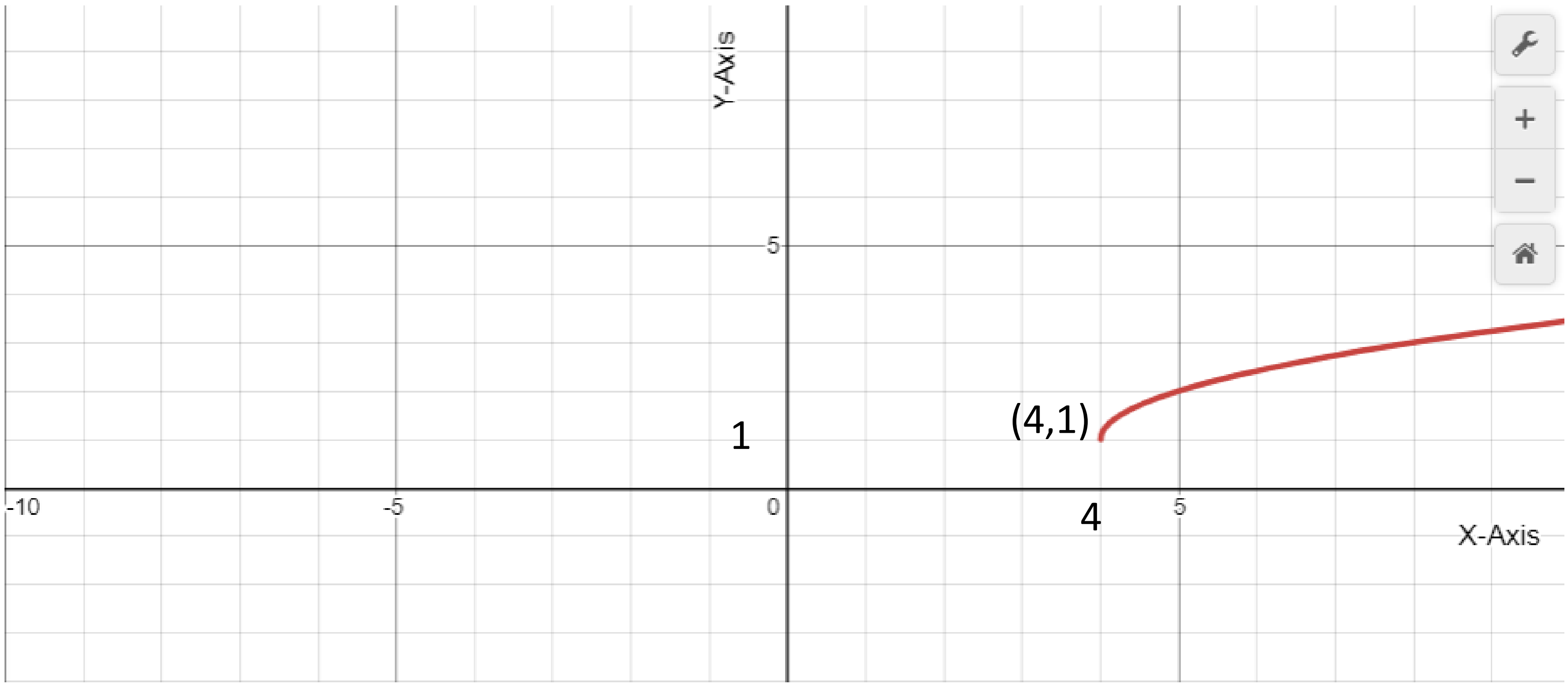




Domain is $[0, +\infty)$ and range is $[0\,, +\infty)$

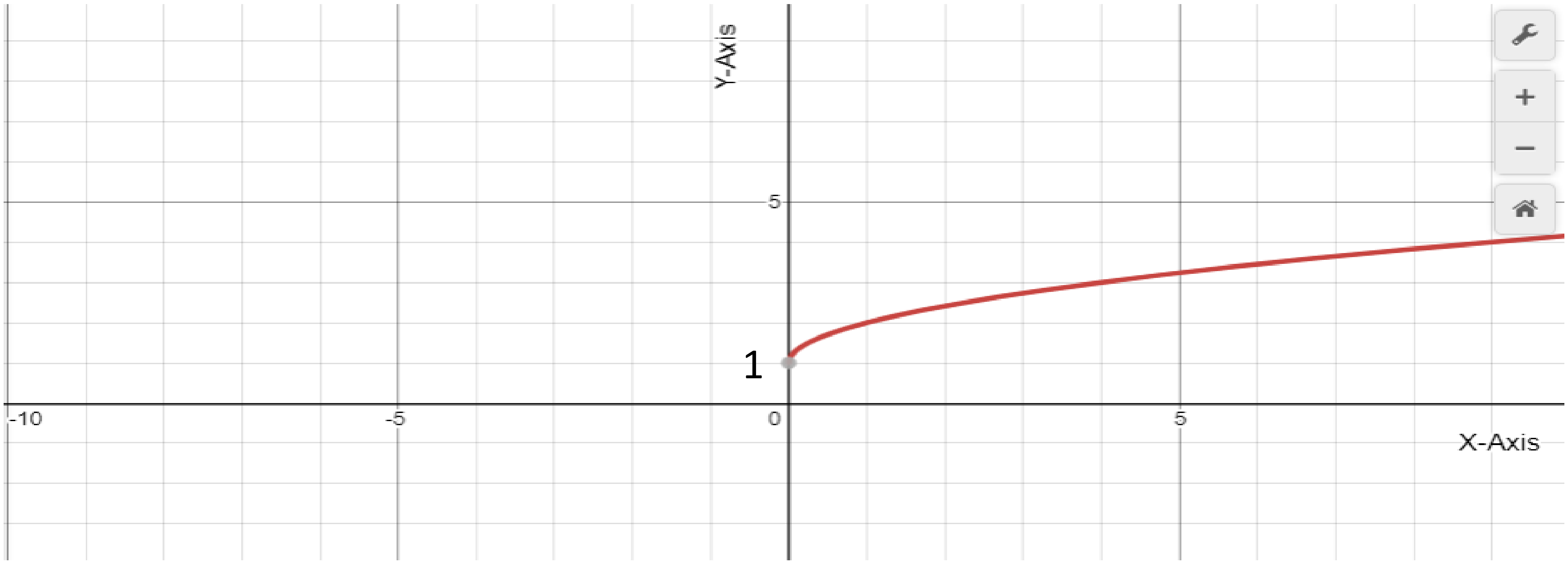
Domain is $[4, +\infty)$ and range is $[0\,, +\infty)$

Graph of $f(x)=y = 1+\sqrt{x-4}$



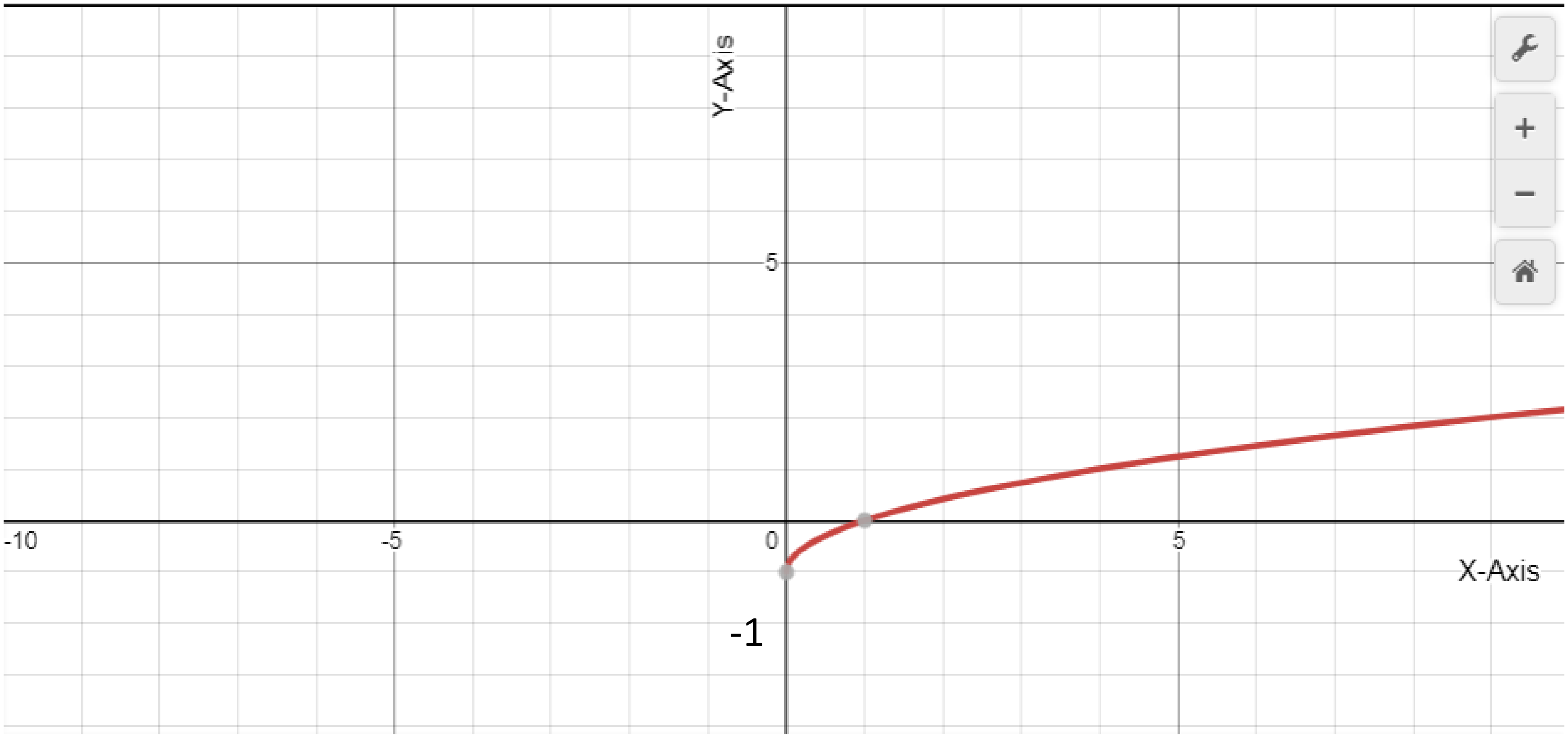
Domain is $[4,+\infty)$ and range is $[1\,,+\infty)$

Sketch the graph of $f(x)=y = 1 + \sqrt{x}$



Domain is $[0 + \infty)$ and range is $[1 , +\infty)$

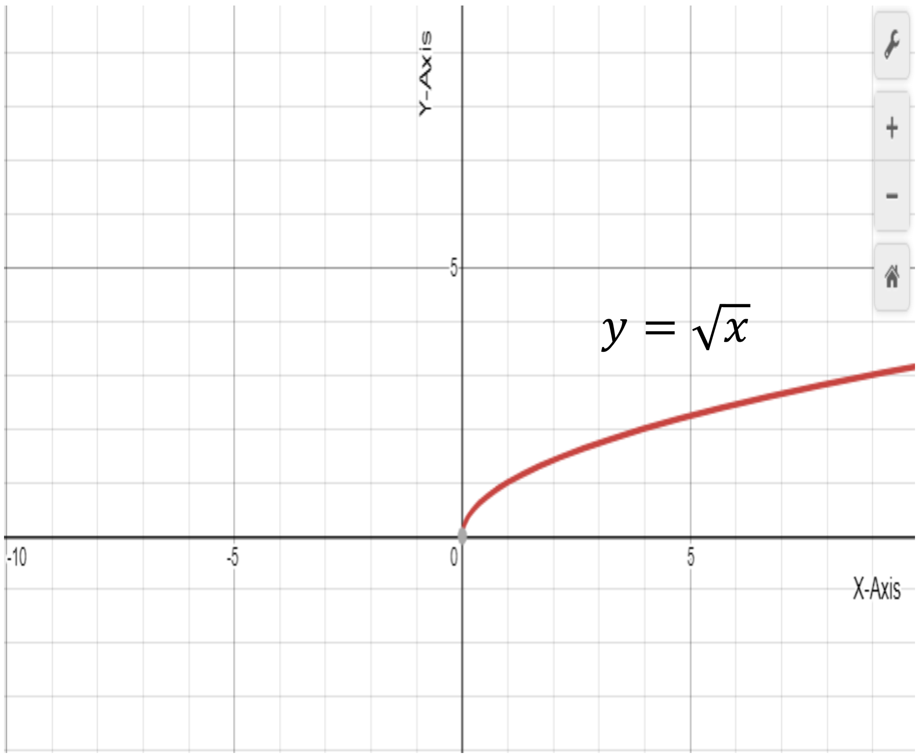
Sketch the graph of $f(x)$=$y = \sqrt{x}$-1

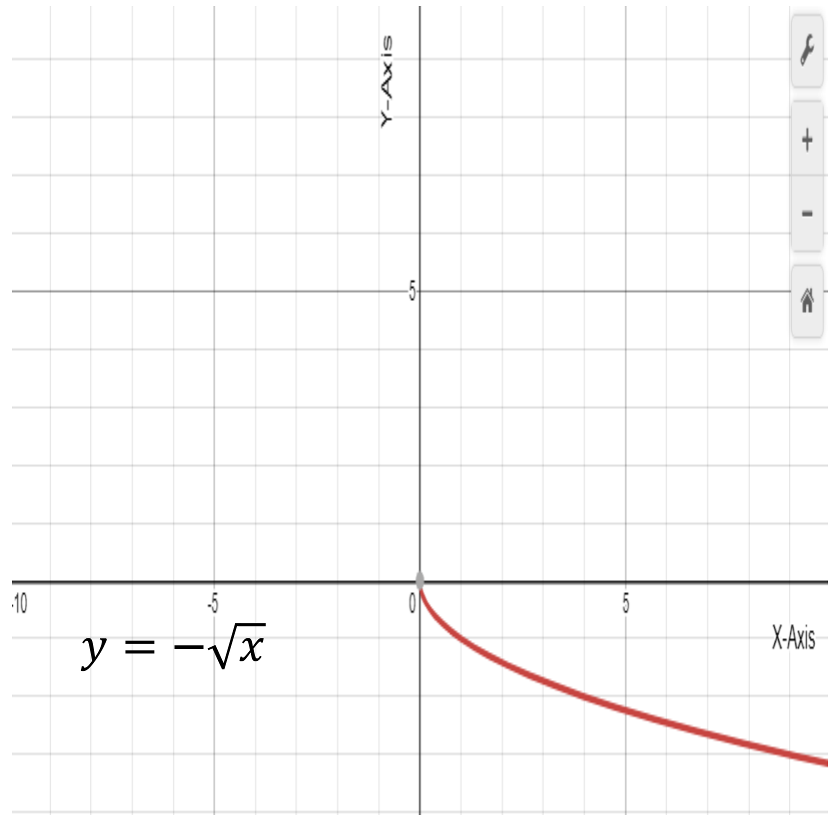


Domain is $[0, +\infty)$ and range is [-1 , $+\infty$)

Exercise: 9(0.2-Page-25):

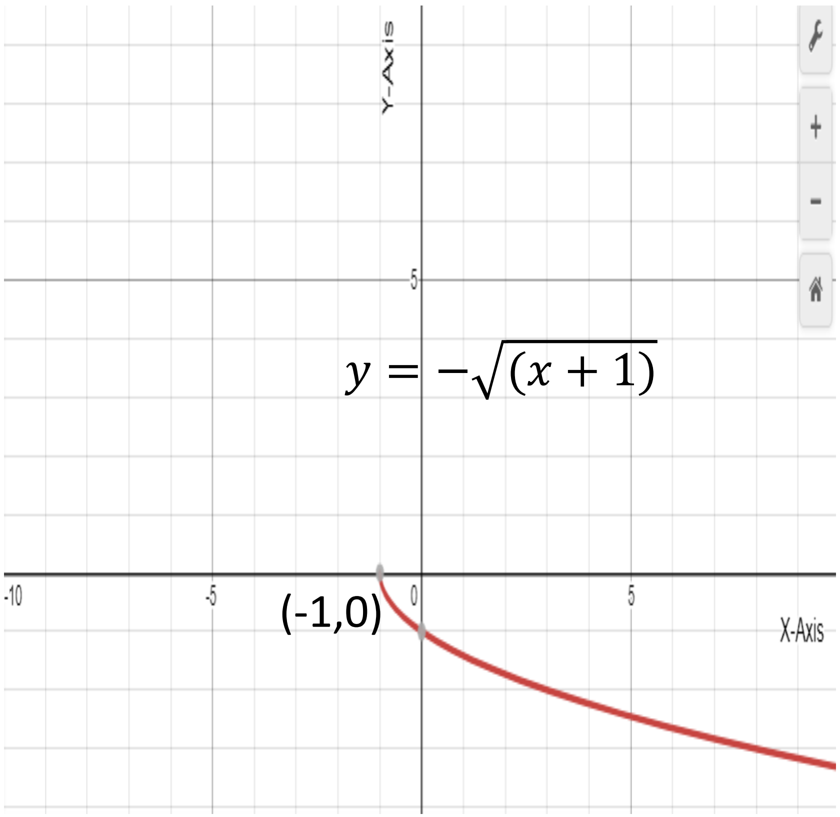
Sketch the graph of $f(x)$=$y = 3 - \sqrt{x+1}$



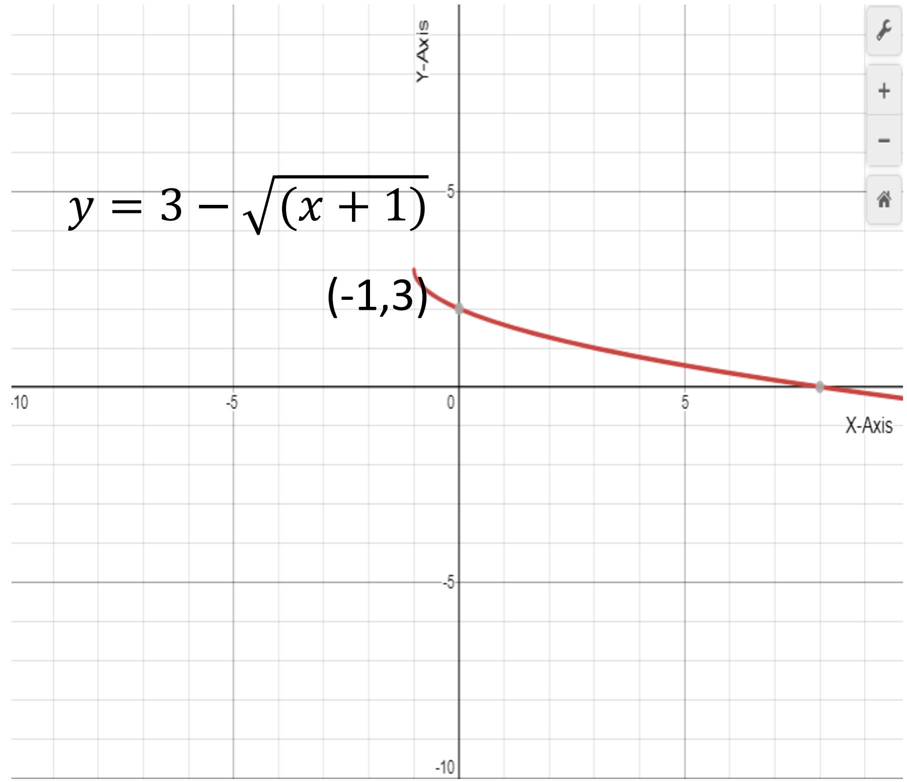


Domain is $[0, +\infty)$ and range is $(-\infty, 0]$

Graph of $f(x)=y = 3 - \sqrt{x+1}$

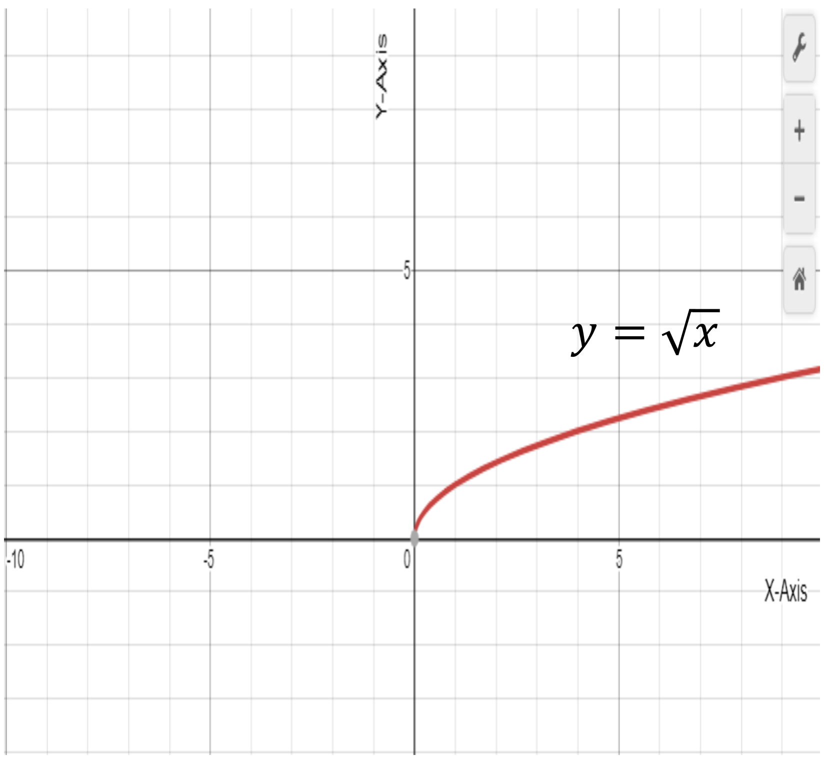


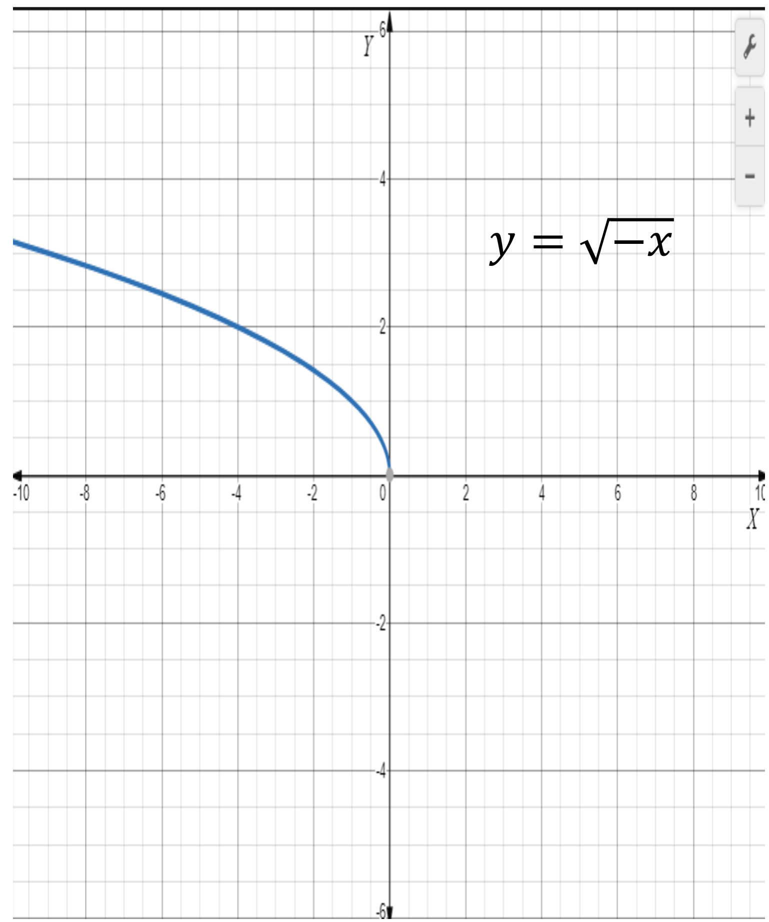
Domain is $[-1, +\infty)$ and range is (-∞, $0]$



Domain is $[-1, +\infty)$ and range is (-∞, $3]$

Sketch the graph of $f(x)$=$y = 3 - \sqrt{-x+1}$

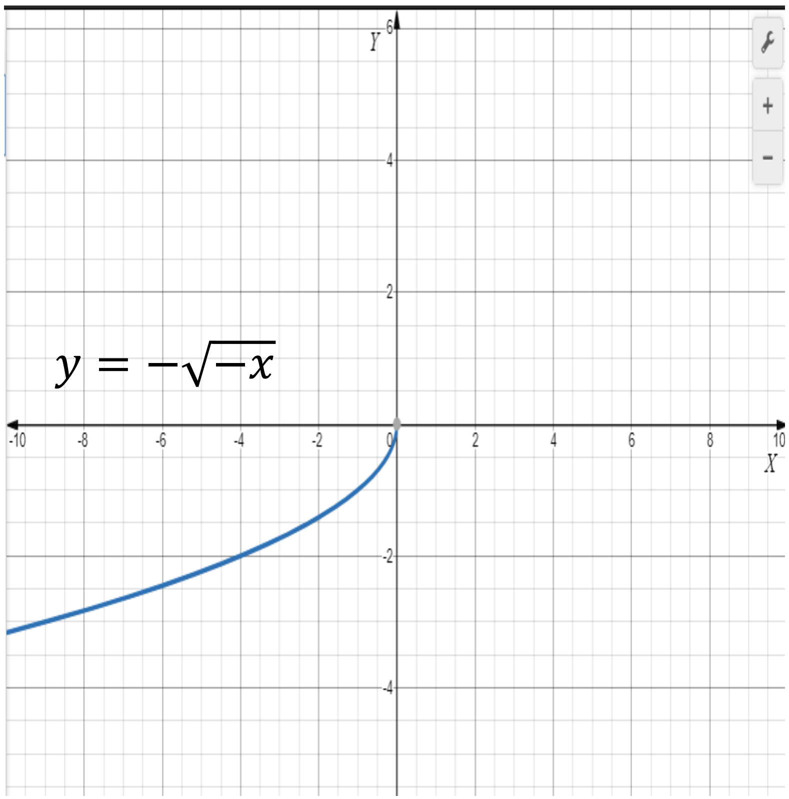


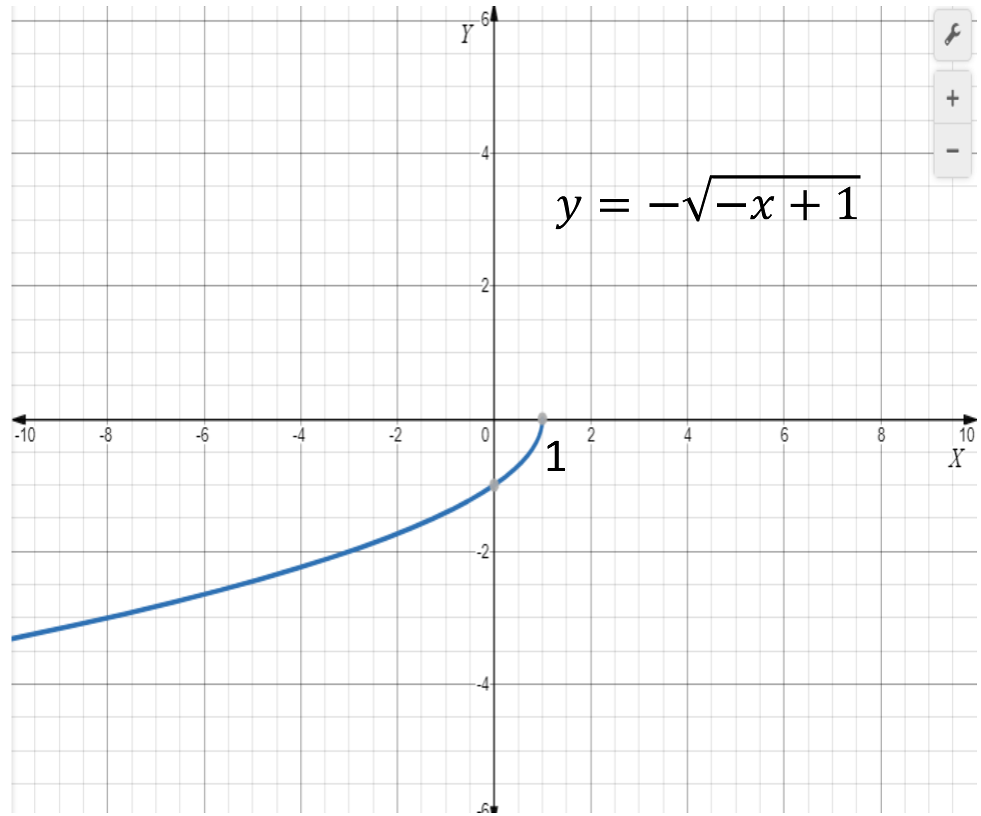


Domain is $(0,+\infty)$ and range is $[0,+\infty)$

Domain is $(-\infty, 0]$ and range is $[0,+\infty)$

Graph of $y = f(x) = -\sqrt{-x+1}$

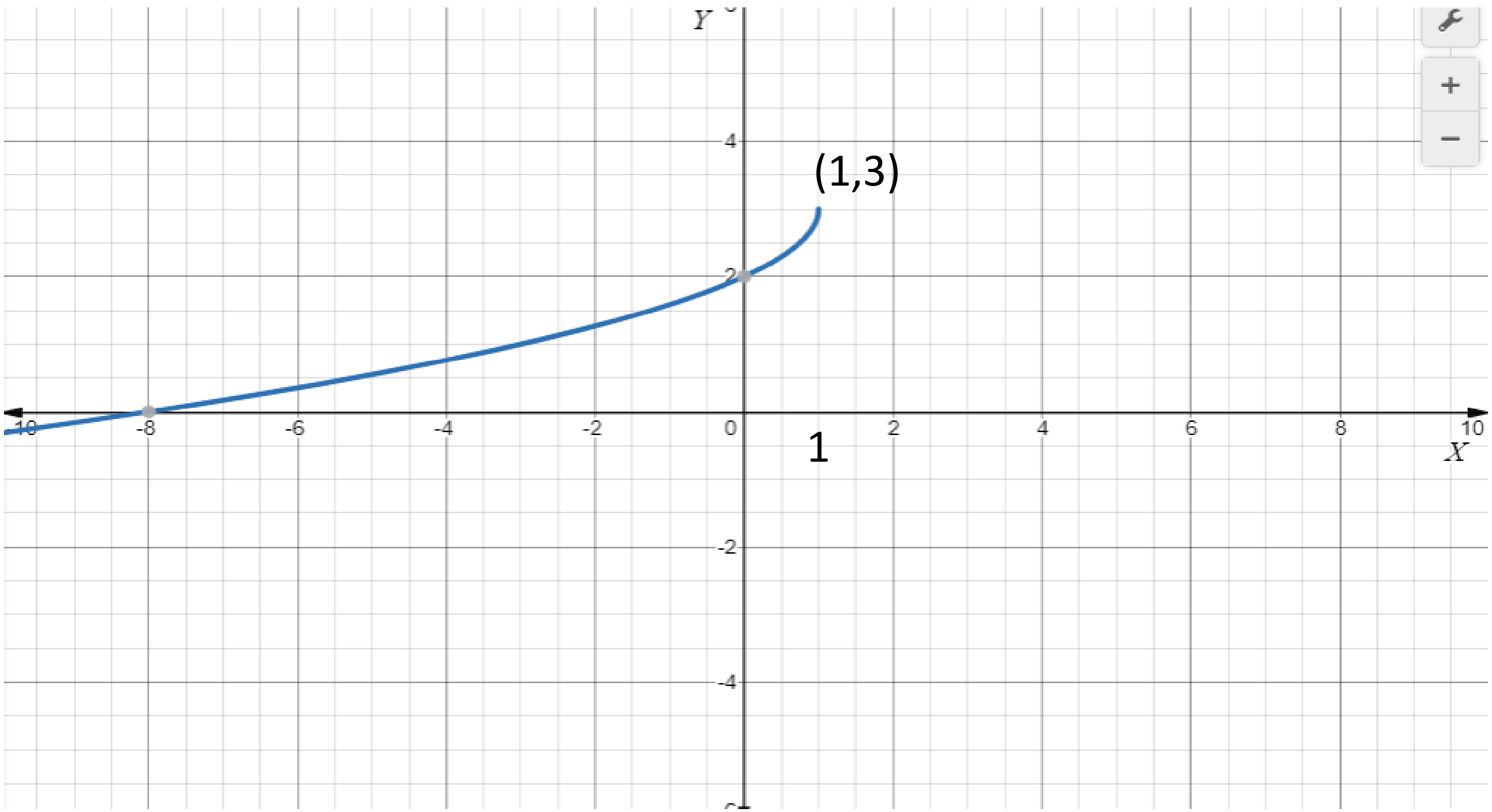




Domain is $(-\infty, 0]$ and range is $(-\infty, 0]$

Domain is $(-\infty, 1]$ and range is $(-\infty, 0]$

Graph of $f(x)$=$y = 3 - \sqrt{-x + 1}$



Domain is $(-\infty, 1]$ and range is $(-\infty, 3]$

Chapter 0.2

- Example 9: Sketch the graph of $f(x) = \sqrt[3]{2-x}$.
-

Solution. Using the translation and reflection principles in Tables 0.2.2 and 0.2.3, we can obtain the graph by a reflection followed by a translation as follows: First reflect the graph of $y = \sqrt[3]{x}$ about the y-axis to obtain the graph of $y = \sqrt[3]{-x}$, then translate this graph right 2 units to obtain the graph of the equation $y = \sqrt[3]{-(x-2)} = \sqrt[3]{2-x}$ (Figure 0.2.5).

Graph of $f(x) = \sqrt[3]{2 - x}$

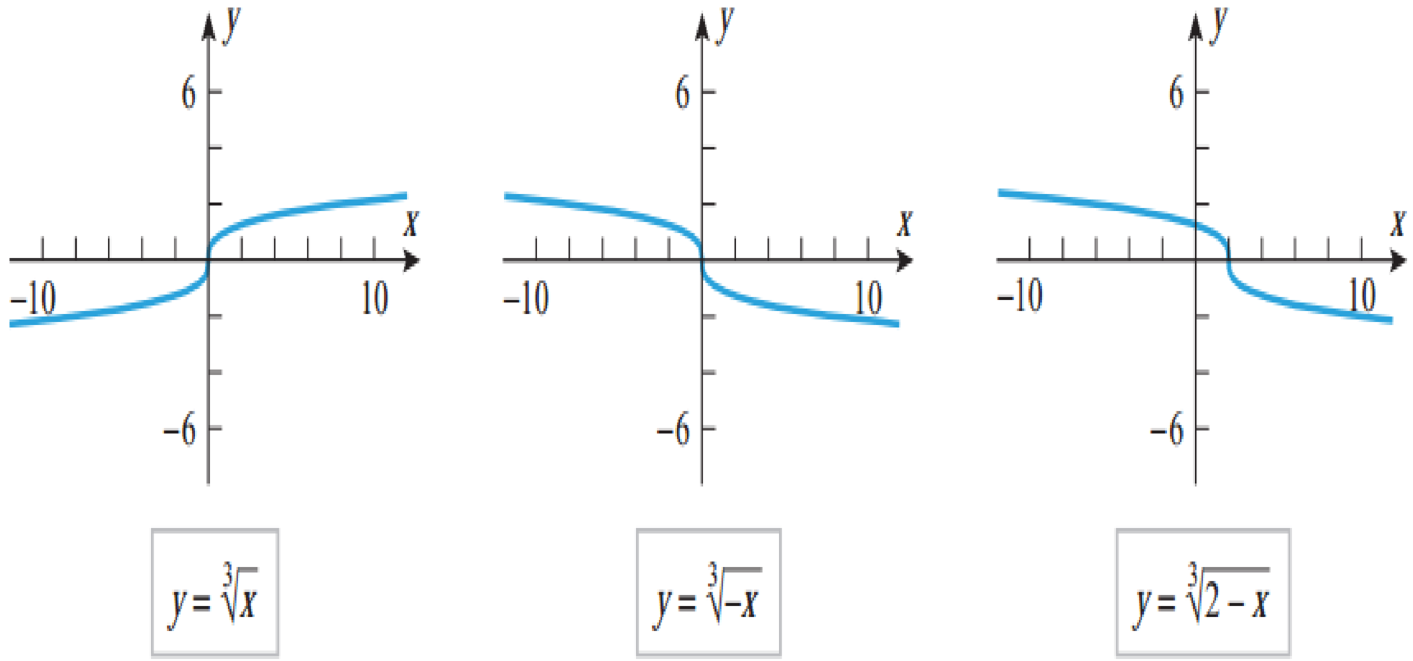
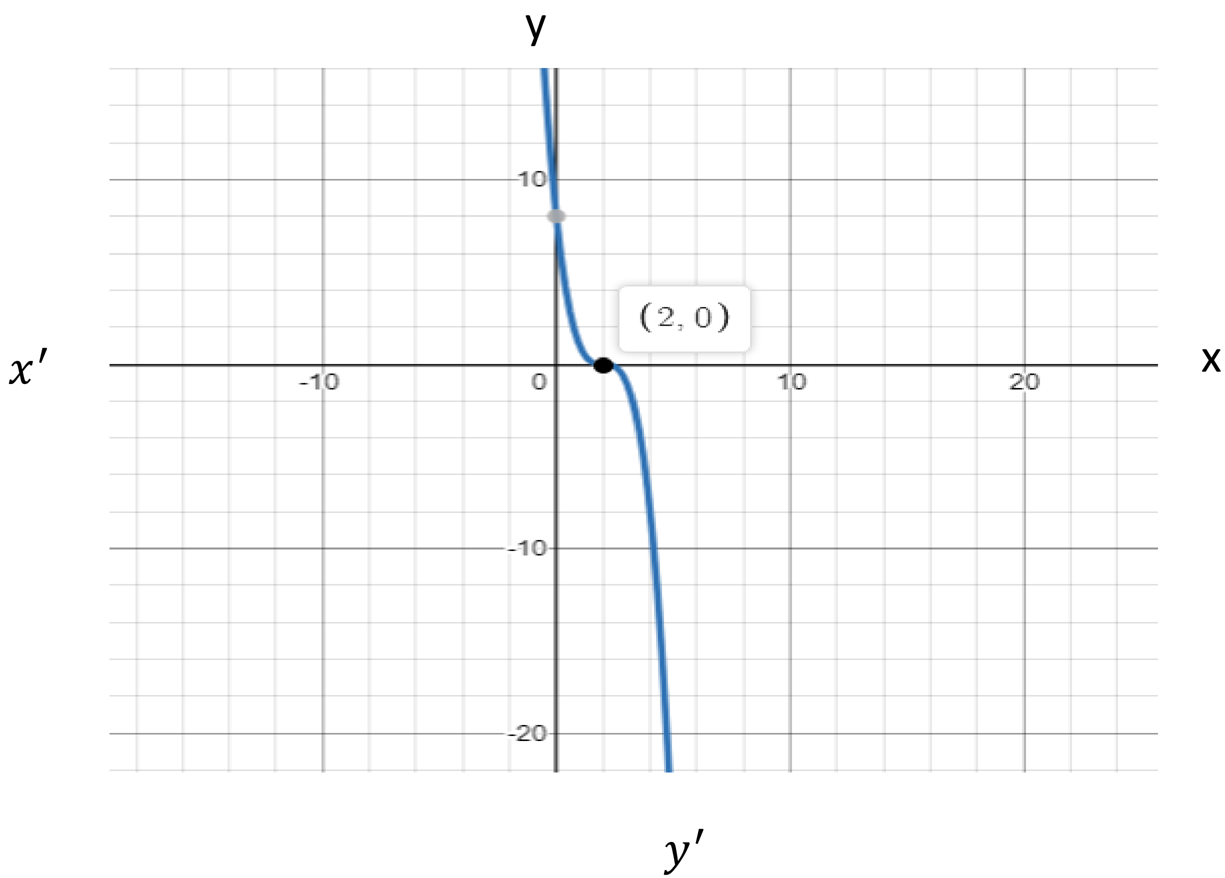


Figure 0.2.5

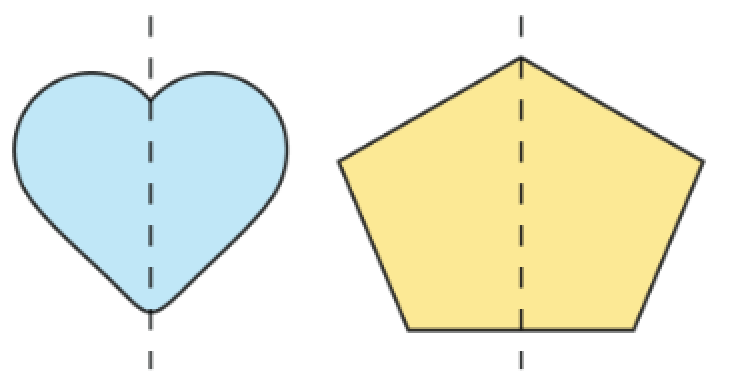
Translations and reflections

- Sketch the graph of $f(x) = (2-x)^3$

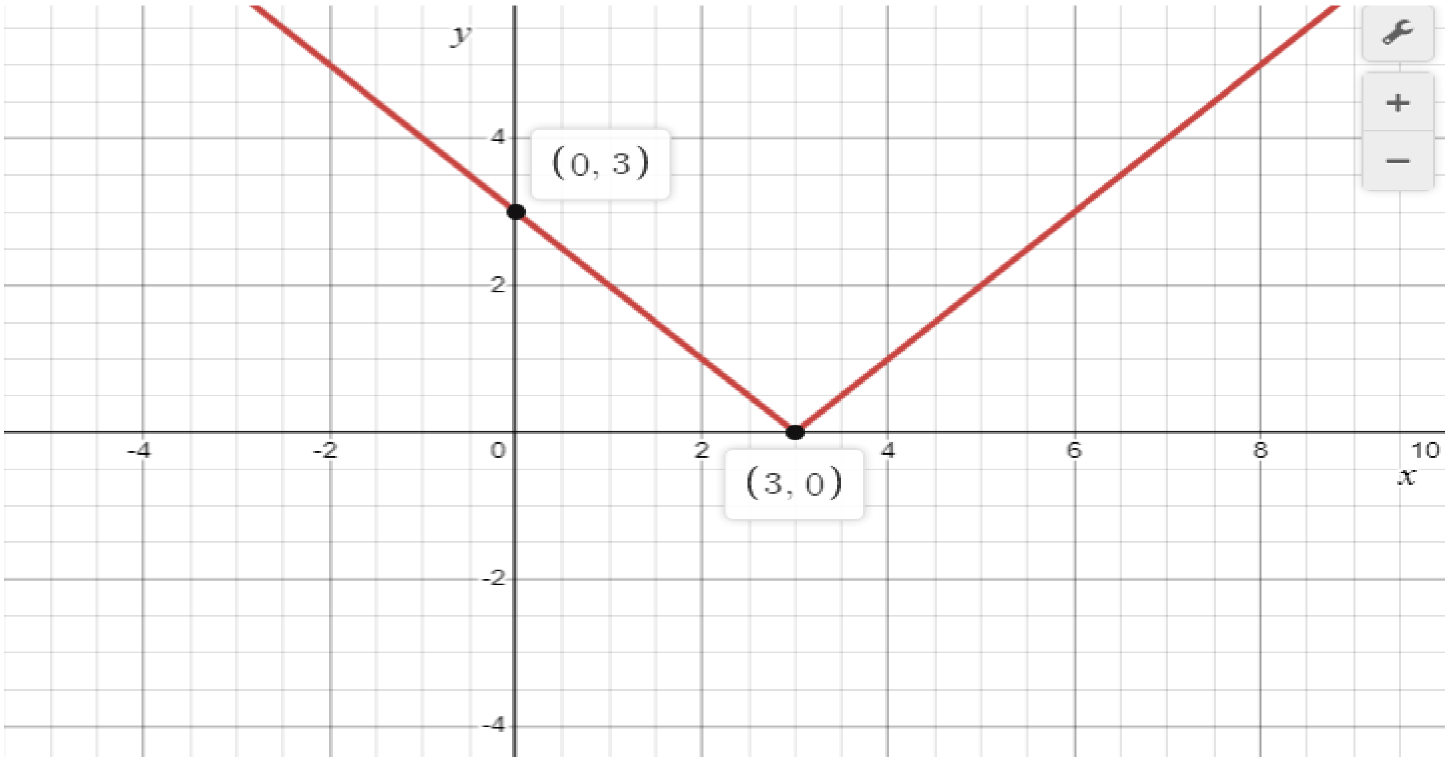


Symmetry

- Symmetry is defined as **a balanced and proportionate similarity that is found in two halves of an object**. It means one-half is the mirror image of the other half.



Sketch the graph of $f(x) = |3 - x|$



Domain is $\mathbb{R}$ $or(-\infty, +\infty)$ and range is $[0\ , +\infty)$